

Model-Driven Software Development

Discussions



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Previous class

Program Analysis

Summary

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Dataflow Analysis

Optimizations with VecDSL



```
program ::= { stmt }

stmt ::= let ID [= expr ]
      | ID = expr
      | print ( expr )
      | if ( expr relop expr )
        { stmt }
        [ else { stmt } ]
      | while ( expr relop expr )
        { stmt }

p ::= == | != | < | <= | > | >=

r ::= expr ( + | - ) expr
      | expr # expr
      | expr ( * | / ) expr
      | - expr
      | expr ->
        ( tpos | len | dim | ID | NUMBER )
      | [ expr { , expr } ]
      | ID | NUMBER | ( expr )
```

Course evaluation

Evaluation

SDU Evaluering

Take some minutes to answer the course evaluation survey

eval.sdu.dk

Evaluation

SDU Evaluering

Should we discuss something now?

Do you want to share something with me/us now?

What is it NOT working?

What is it working (and should continue doing)?

Extra comments

Me / Lectures / Discussions / Tools / Polls / Assignments / Slides / in-class exercises / Discussion board for exercises / Semester project / ...

Model-Driven Software Development

Discussions



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Exam

Exam

MCQ: ~10 questions / 0.5 hours

- 1 point per question

- Multiple answers: choose all right answers

- No *programs* questions, but there can be code as example

- No aids, no internet

Open-ended questions: 4 parts / 2.5 hours

- Hand-in: PDF with answers and (optional) individual extension code

- Aids: Slides, source code of your project, notes, etc.

- No internet, no gen AI

- Questions *heavily* based on your project and individual extension

Exam: MCQ

Content: 12 lectures

Model-driven, DSL and PL, Grammar and Xtext, PX, Types, Analysis ...

No Code

But, code can be used as an example

Xtext grammar to define Metamodel

DSLs to define AST

Exam: open-ended

- 3 Parts on Group Project

 - Each part is divided by a general topic

- 1 Part on Individual Extension

- Parts are not distributed *equally*

 - The exam is divided in terms of content, not %

- 3 Parts $\neq$ 3 Questions

 - Each part has several questions

Exam: open-ended – Question Example

Question 1

What are the critical concepts/properties you model from The Problem?
Why do you think these concepts/properties are important?

Expected answer:

Describe the properties you include during the solution design of The Problem.
Justify the importance of these properties in your design.

Exam: open-ended – Question Example

Question 1

What are the critical concepts/properties you model from The Problem?

Why do you think these concepts/properties are important?

Expected answer:

Describe the properties you include during the solution design of The Problem.

Justify the importance of these properties in your design.

Answer → itslearning plan *Project: Metamodeling*

Exam: open-ended – itslearning

We use itslearning as a way for you to practice for the exam

Some (most?) of the questions will be based on itslearning project questions

However, your answers are (probably) outdated

I recommend you to go back to the questions and study

We **will not give** any more feedback in itslearning

Even when questions may be based on itslearning, they still can ask for more

Project properties – in itslearning

Properties importance – not in itslearning

Exam: open-ended – part 0

I will send (soon) a part 0

This part is not evaluated, but it is here to give context to your answers

Part 0

Group Topic to answer the exam: e.g., Class Description

Short description: e.g., DSL to describe class in a OOP PL

Part 4 – Question 0:

Topic of Extension: e.g., Type, Validation, Testing, PX, ...

Miscellaneous Metamodel

Model vs metamodel vs meta metamodel ...

Program: *concrete* syntactical object in a given language

Model: abstraction over similar programs

Metamodel: abstraction over similar models (e.g., from a domain)

Metametamodel: metamodel of metamodels (e.g., xtext grammar language)

Meta... : UML

Meta... : UML

...

Meta is relative

Meta is relative: the case of Pharo Smalltalk

Everything is an object

Every object is an instance of a Class

Every class has a super class

Classes are objects too!

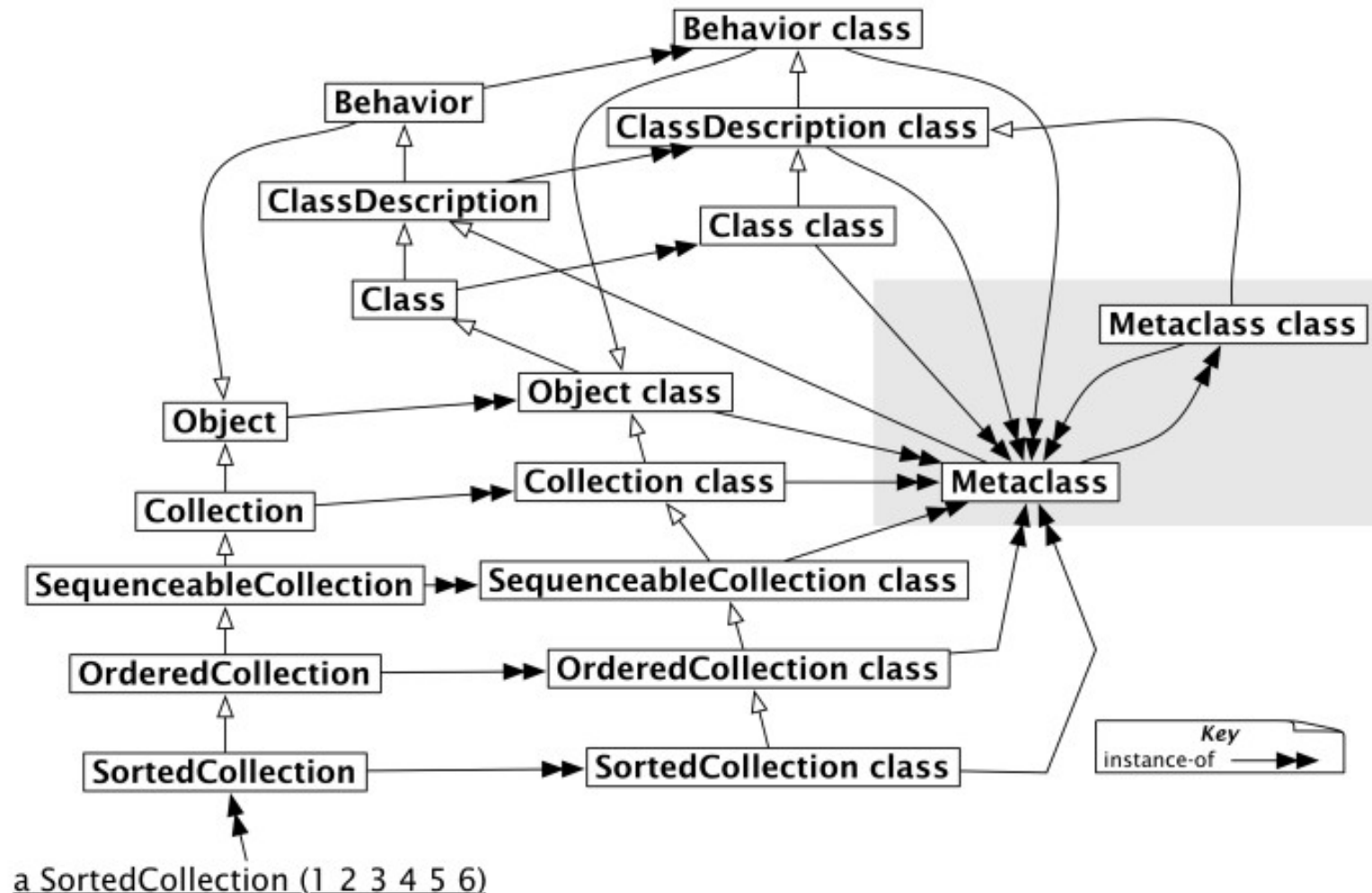
What is the class of an object?

What is the class of a class?

What is the class of a class of a class?

What is the class of a class of a class of a ... ?

Meta is relative: the case of Pharo Smalltalk



Miscellaneous

Programming languages

Programming style: Imperative vs Declarative

Imperative

Program instructs computer how to compute result

Classical imperative GPLs such as Java and Python (but: class hierarchy)

Precise control at the cost of abstraction

Your classic PL

Your DSL?

Declarative

Program specifies what the result should be

Many DSLs but also GPLs such as Prolog (functional languages in between)

Abstraction at the cost of loss of control

OOP (if well done), SQL

Your DSL?

MDSD/Xtext vs compilers

MDSD course: *compiler course light* + software engineering + DSL magic

Compilers

Lexing and parsing produced an AST

Name resolution phase transforms AST to *metamodel instance*

Multiple additional phases (validation, desugaring, optimization, etc)

Final phase produces output (binary code, assembly, C, Java bytecode, etc)

Model-aware vs model-ignorant generation

Model-aware

```
State a = new State()
a.onExecute{//lamda with code}
State b = new State()
...
t = new Transition(a, b, 'event');
...
Machine m = new Machine();
m.addState(...)...
m.addTransition(...)...
m.execute()
```

Model-ignorant

```
public void processEvent(String event){
    switch(currentState){ //on state
        case 0: //if in state 0
            switch(event) { //events
                case 'event': {
                    //process code
                    //next state if applicable
                }
            }
        case 1: //if in state 1 ...
    }
}
```

Code generation: template vs transformation

Template-based

```
def generateJavaString(MathExp math) '''
package math_expression;

public class «math.name.toFirstUpper»{

    «FOR varBinding: math.variables»
    public int «varBinding.name»;
    «ENDFOR»

    «IF !math.externals.empty»
    public interface External {
        «FOR external: math.externals»
        «external.generateString»;
        «ENDFOR»
    }

    private External externalMath;

    public «math.name.toFirstUpper»(External external) {
        externalMath = external;
    }
}
```

Transformed-base

```
def String generateExpressionString(Expression exp){
    switch exp{

        Plus: '''(«exp.left.generateExpressionString» + «exp.right.generateExpressionString）」'''
        Minus: '''(«exp.left.generateExpressionString» - «exp.right.generateExpressionString）」'''
        Mult: '''(«exp.left.generateExpressionString» * «exp.right.generateExpressionString）」'''
        Div: '''(«exp.left.generateExpressionString» / «exp.right.generateExpressionString）」'''
    }
}
```

MSc Topics

MSc Topics:

General

MDSD – DSLs – SE for Robots

Particular

Programming Experience for robotic/drone developers

MDSD for robotic/drone developers

Misc

MSc Topics: PX for robots

- Live programming of <robot> applications

 - Mobile, Cobots, Drones, Soft Robots, CPS, ...

- Techniques for understanding and creating programs for robots

 - Extended Reality,...

- Research and improve debugging practices of (r/c)obots applications

 - Classic* debugging

 - Omniscient debugging of (r/c)obots applications

 - Pharo-based debugging for robots

 - Simulation re-play of robotic executions

MSc Topics: PX for robots - Concrete

Pharo-based live programming for Cobots

Re-programmability of Cobots using custom IDE

VR for robotic visualizations within simulated environments

Pharo-based IDE for live programming for <robots>

More topics will come...

MSc Topics: MDSD for robots

Programming Languages for <robot>

Mobile, Cobots, Drones, Soft Robots, CPS, ...

Live (re)planning of multi-drone missions

DSL for multi-drone missions in simulation (for hundred of drones)

DSL for semantic databases

Software Architecture for robot research

More topics will come...

MSc Topics: Misc

Green Topics – new for me

- Software for Sustainability and Communities

 - DBA but *better*

 - Why local shops have to be in google maps?

- Software without colonialism (no teams, google, etc.)

- Need scope and refinement

Education

- How to help you learn DSLs?

- How to help you learn Web Development?

- How to help me to teach you better? (tools, practices, style, ...)

Final words

Final words

Models are part of the development process

They are equal to code

We can *generate* code from models

We can *transform* models to other models

Domain is essential for model-driven systems

We build models using *metamodels*

It is ok to not use DSL in the future, but...

Do not discard them right away!

Metamodeling is still key for your career

Be curious and keep learning...